

Exercises about regular expressions

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Question 1. Identify the lexemes and their corresponding token in the following C program.

```
/* Return maximum of integers i and j */  
int max (int i, int j) {  
    return i>j? i : j;  
}
```

Question 2. Let the alphabet $\Sigma = \{a, b\}$ and the following regular expressions:

$$r = a(a|b)^*ba$$

$$s = (ab)^*|(ba)^*|(a^*|b^*)$$

The language denoted by r is noted $L(r)$ and the language denoted by s is noted $L(s)$.

Find a word x such as

1. $x \in L(r)$ and $x \notin L(s)$
2. $x \notin L(r)$ and $x \in L(s)$
3. $x \in L(r)$ and $x \in L(s)$
4. $x \notin L(r)$ and $x \notin L(s)$

Question 3. Given the binary alphabet $\Sigma = \{a, b\}$ and the order on letters $a < b$, write regular definitions for the following languages.

1. All words starting and ending with a .
2. All non-empty words.
3. All words in which the third last letter is a .
4. All words containing exactly three a .
5. All words containing at least one a before a b .
6. All words in which the letters are in increasing order.
7. All words with no letter following the same one.

Question 3bis. Let the alphabet be now the ASCII. Find regular definitions for

1. All comments on one line between braces, i.e., { and }.
2. All comments between /* and */ without */ inside, except between double-quotes, i.e. ".

Hint: Use the complement of a letter in the alphabet, e.g., $\overline{A}$ denotes all letters except A. g

Question 4. Simplify, if possible, the following regular expressions.

$$\begin{aligned} &(\epsilon | a^* | b^* | a | b)^* \\ &a(a | b)^*b | (ab)^* | (ba)^* \end{aligned}$$